Physical Chemistry (I)

Lecture 13

Phases of Pure Substances

In the Last Class

- Classical statistical mechanics
  - Phase space
  - Calculation of partition function
- Applications
  - Perfect gas
  - Real gas
  - Equipartition theorem

Phase Space

Space of all possible microstates of a system

- Classical mechanics:

Space of all possible values of

position and momentum variables

Last class

Classical Partition Function

$$q = \sum_s e^{-E(s)/k_\mathrm{B}T} \quad \rightarrow \quad q = \frac{1}{h} \int \int e^{-E(x,p)/k_\mathrm{B}T}\, \mathrm{d}x \mathrm{d}p$$

for a 1-dimensional single particle

Last class

Calculation of Q

For indistinguishable particles in 3 dimensions,

$$Q = \frac{1}{N!\,h^{3N}}\left[\int\int e^{-E(\vec{r},\vec{p})/k_\mathrm{B}T}\,\mathrm{d}\vec{r}\mathrm{d}\vec{p}\right]^N = \frac{1}{N!}q^N$$

where

$$q = \frac{1}{h^3}\int\int e^{-E(\vec{r},\vec{p})/k_\mathrm{B}T}\,\mathrm{d}\vec{r}\mathrm{d}\vec{p}$$

Perfect Gas

- No interactions (no potential energy)
- Total energy = kinetic energy $(p^2/2m)$
- Bound in a box of volume V
  - Range of x = $(0, a)$
  - Range of y = $(0, b)$
  - Range of z = $(0, c)$
  - $abc = V$
- Range of momentum = $(-\infty, \infty)$

Q of Perfect Gas

$$q = V\left(\frac{2\pi m k_{\mathrm{B}} T}{h^2}\right)^{3/2}$$

For indistinguishable gas particles,

$$Q = \frac{1}{N!} q^N = \frac{V^N}{N!}\left(\frac{2\pi m k_{\mathrm{B}} T}{h^2}\right)^{3N/2}$$

Thermodynamics

$$Q = \frac{V^N}{N!}\left(\frac{2\pi m k_\mathrm{B} T}{h^2}\right)^{3N/2}$$

$$A = -Nk_\mathrm{B}T\ln V + k_\mathrm{B}T\ln N! - \frac{3}{2}Nk_B T\ln\left(\frac{2\pi m k_\mathrm{B} T}{h^2}\right)$$

$$U = \frac{3}{2}Nk_\mathrm{B}T$$

$$p = \frac{Nk_\mathrm{B}T}{V}$$

Last class

Pressure of Real Gas

Partition function

↓

Helmholtz energy

↓

Pressure

$$p = \frac{nRT}{V - nb} - a\frac{n^2}{V^2}$$

Last class

Quadratic Contributions

$$Q = \frac{1}{N!\,h^{3N}} \int \cdots \int e^{-E_{\mathrm{tot}}/k_{\mathrm{B}}T}\,\mathrm{d}\vec{r}_1\mathrm{d}\vec{p}_1\mathrm{d}\vec{r}_2\mathrm{d}\vec{p}_2 \cdots \mathrm{d}\vec{r}_N\mathrm{d}\vec{p}_N$$

Assume that

$$E_{\mathrm{tot}} = \sum_{i=1}^{f_r} A_i r_i^2 + \sum_{j=1}^{f_p} B_j p_j^2$$

$$(A_i, B_j > 0)$$

and that the ranges of both positions and momenta are $(-\infty, \infty)$

Equipartition Theorem

$$Q = \frac{\pi^{(f_r+f_p)/2}}{N!\,h^{3N}}\left(\frac{1}{\beta}\right)^{(f_r+f_p)/2}\prod_{i=1}^{f_r}\left(\frac{1}{A_i}\right)^{1/2}\prod_{j=1}^{f_p}\left(\frac{1}{B_j}\right)^{1/2}$$

$$U = \left(f_r + f_p\right)\frac{k_{\mathrm{B}}T}{2}$$

"The average value of each quadratic contribution

to the internal energy is $(1/2)k_{\mathrm{B}}T$."

Where Are We?

Foundations

Basic concepts

Gases

Thermodynamics

Statistical mechanics

Applications

Phases

Simple mixtures

Chemical equilibria

7. Phase Transitions

Chapter Overview

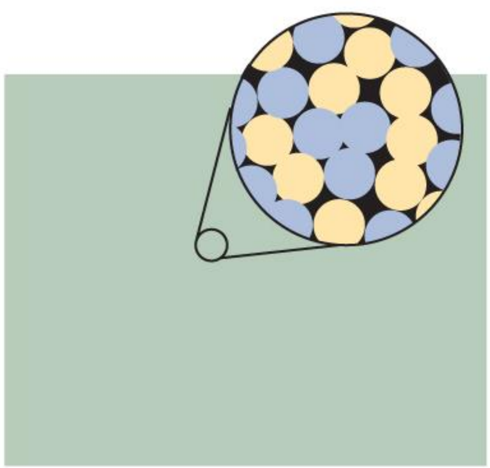
- Phase and chemical potential
  - Phase and phase diagram
  - Chemical potential
  - Phase rule
- van der Waals fluid
  - Liquid-gas phase transition
  - Supercritical fluid
  - Critical point and the principle of corresponding states
- Thermodynamic aspects of phase transitions

Phase

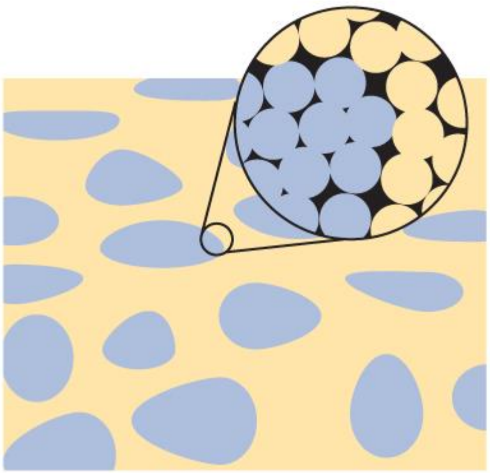
A form of matter that is uniform throughout in chemical composition and physical state

Topic 4A

Solution: Single Phase



solution
(one phase)



dispersion
(two phases)

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figures 4A.1)

Phase Diagram

A diagram that shows the regions at which its various phases are thermodynamically stable

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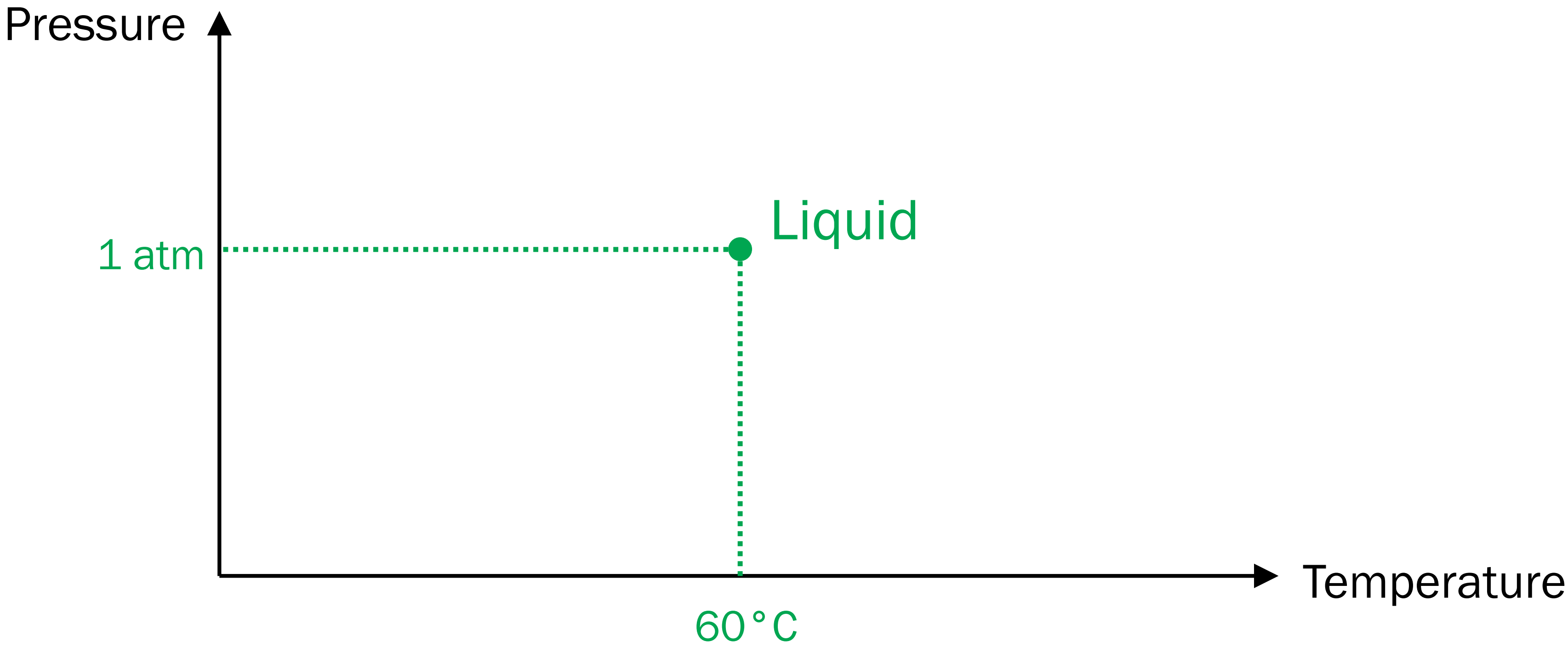
Drawing Phase Diagram



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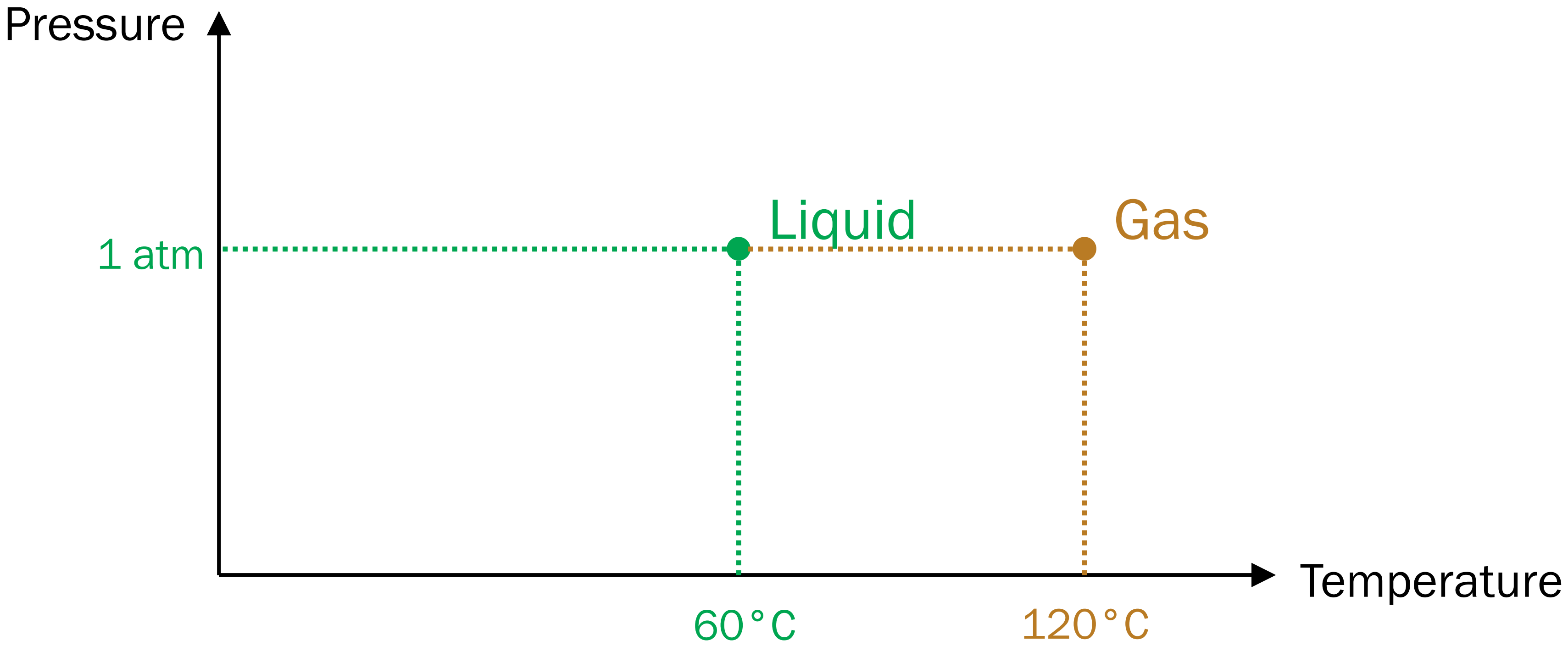
Drawing Phase Diagram



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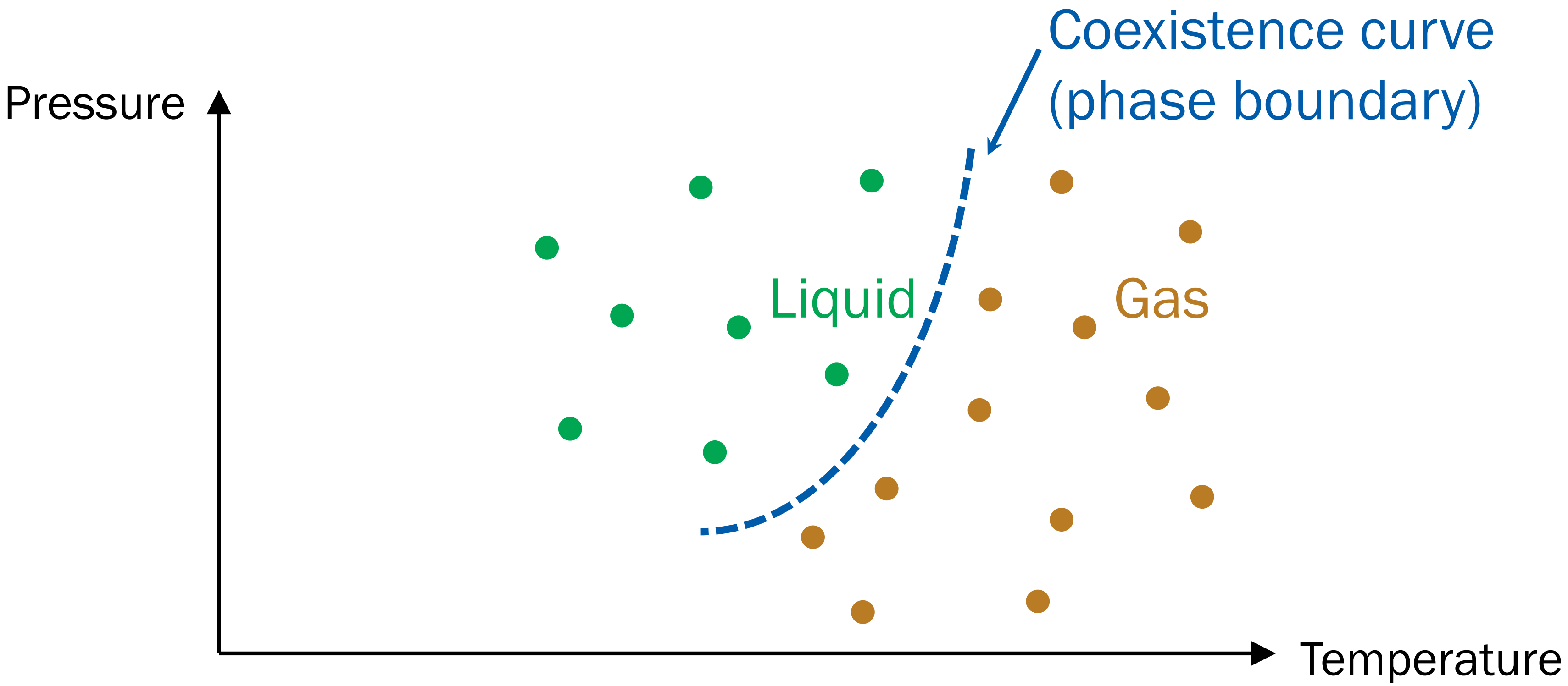
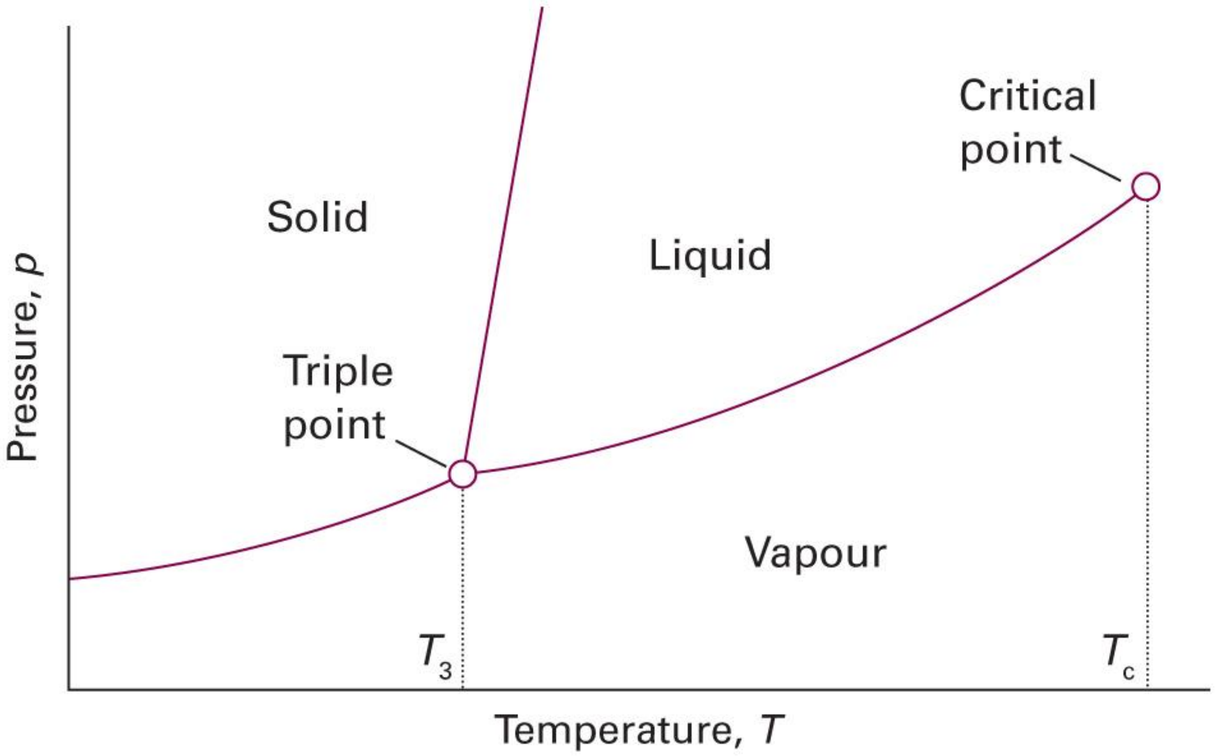
Drawing Phase Diagram



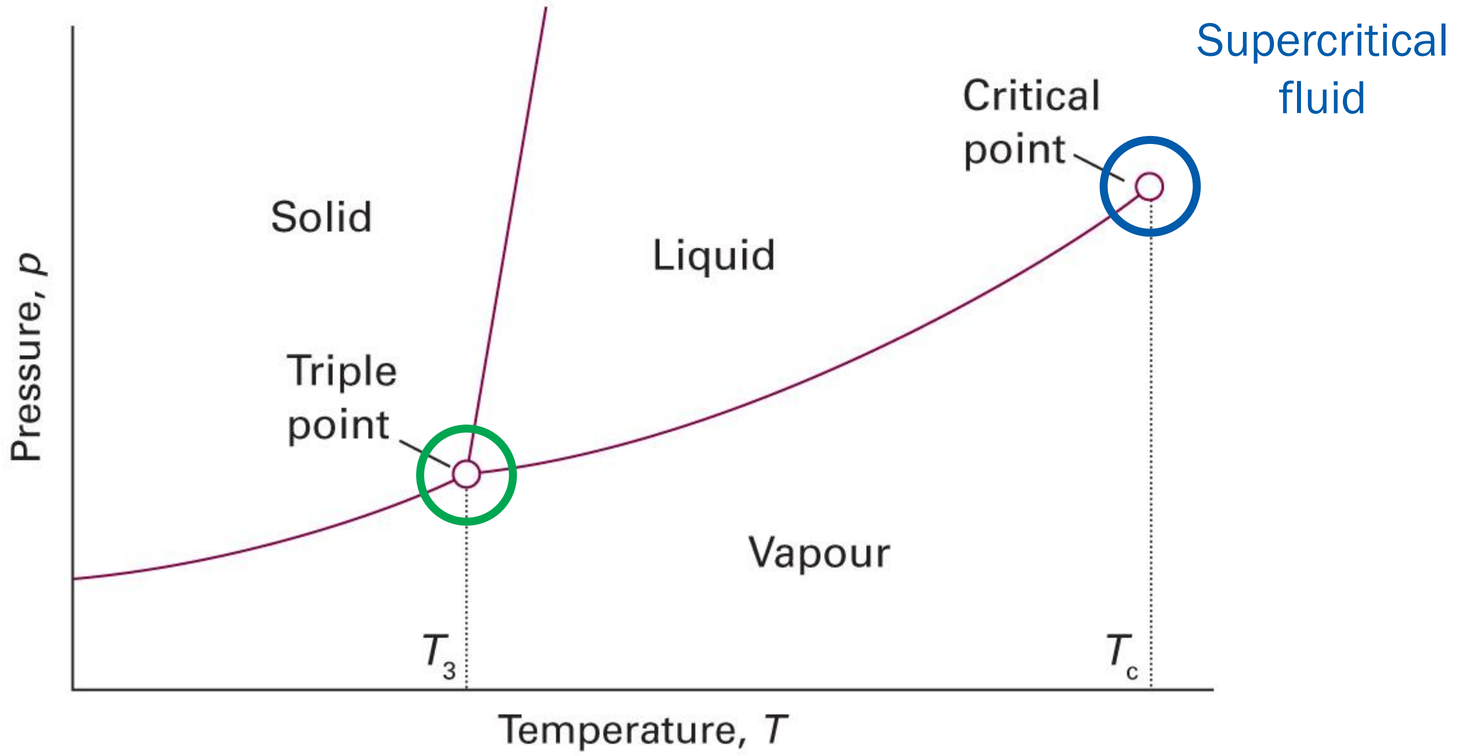
Topic 4A

Typical Phase Diagram



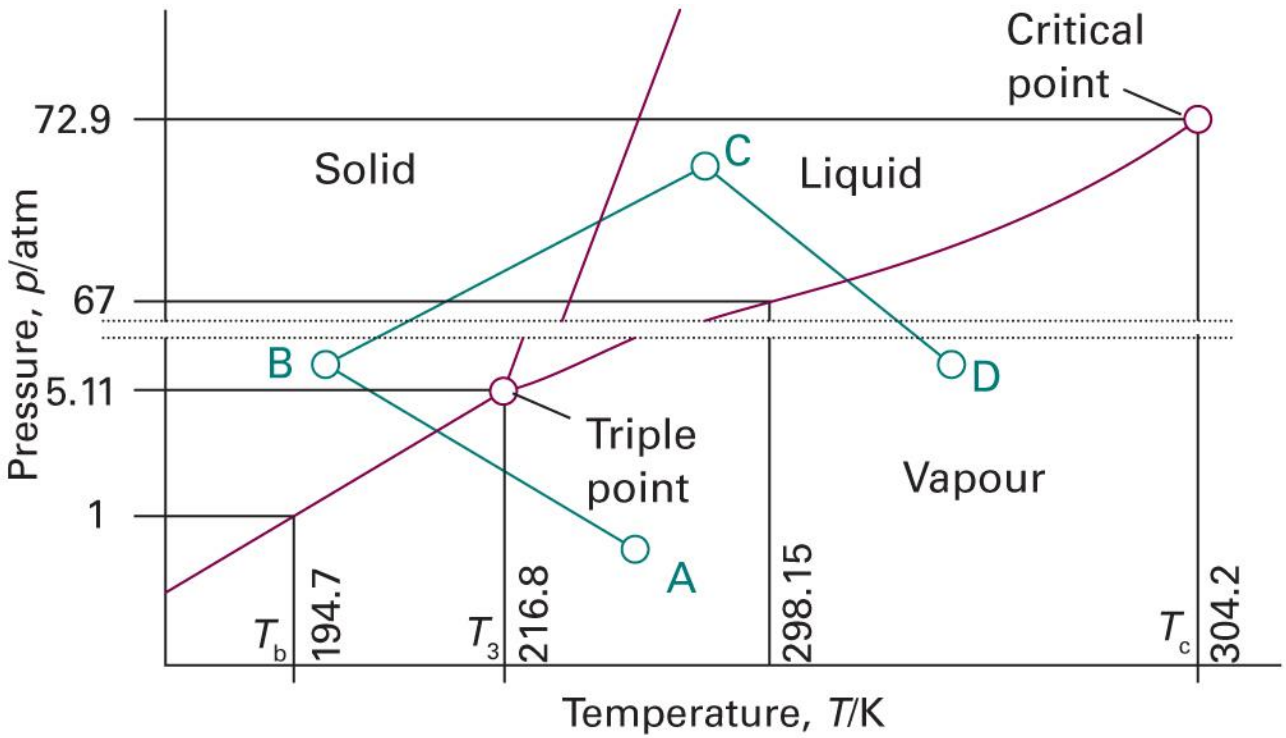
(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figures 4A.4)

Two Characteristic Points



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figures 4A.4)

Carbon Dioxide



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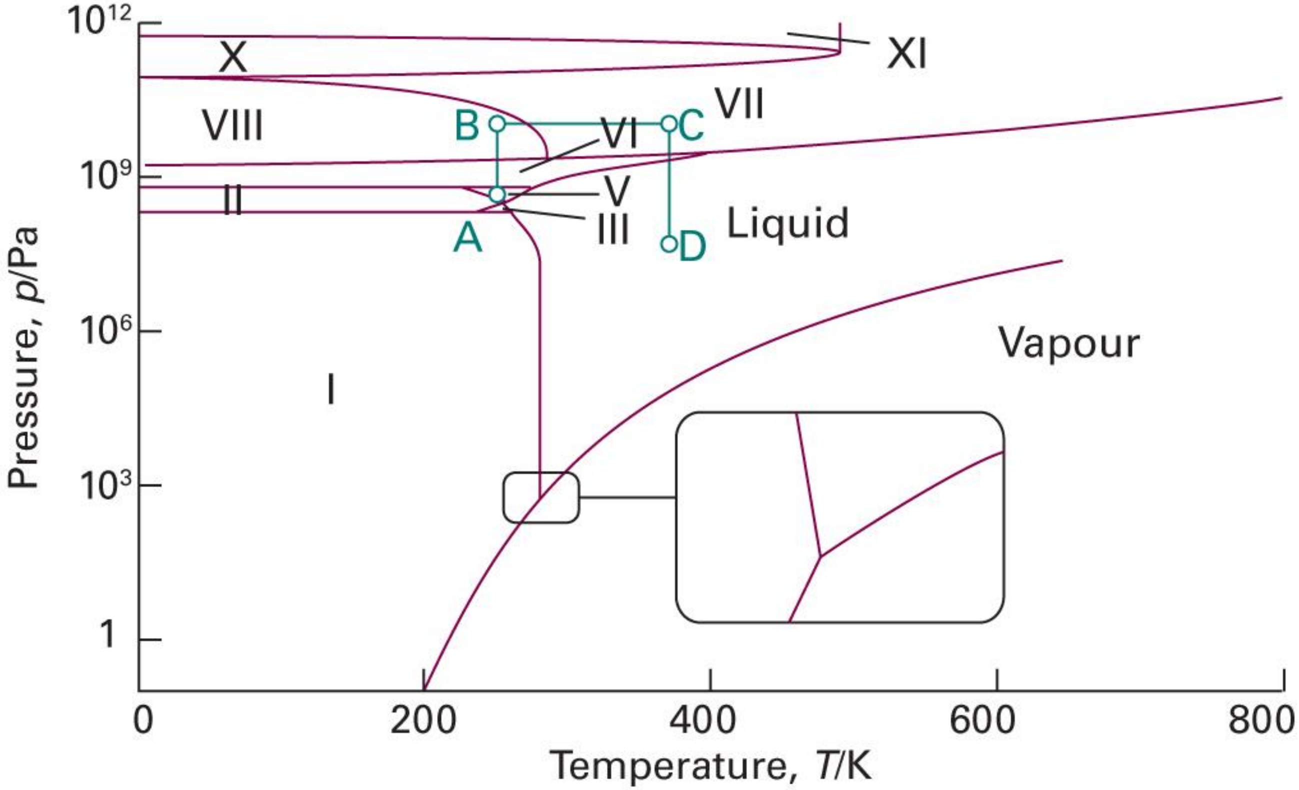
Water



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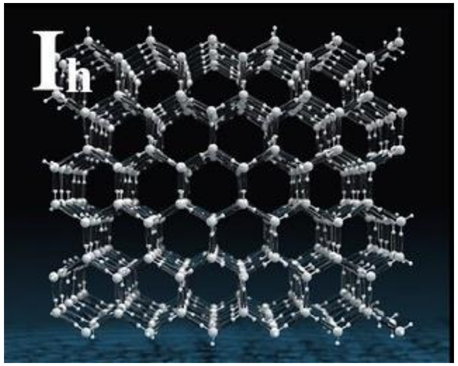
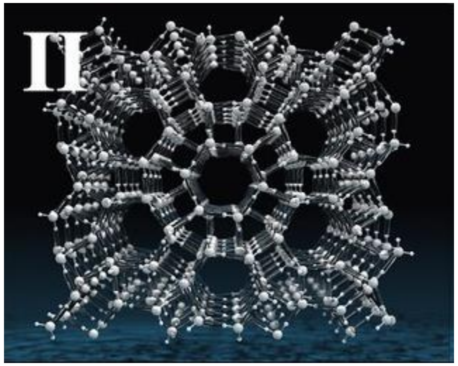
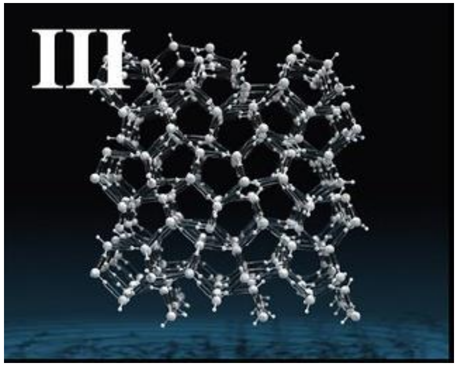
Different Phases of Ice

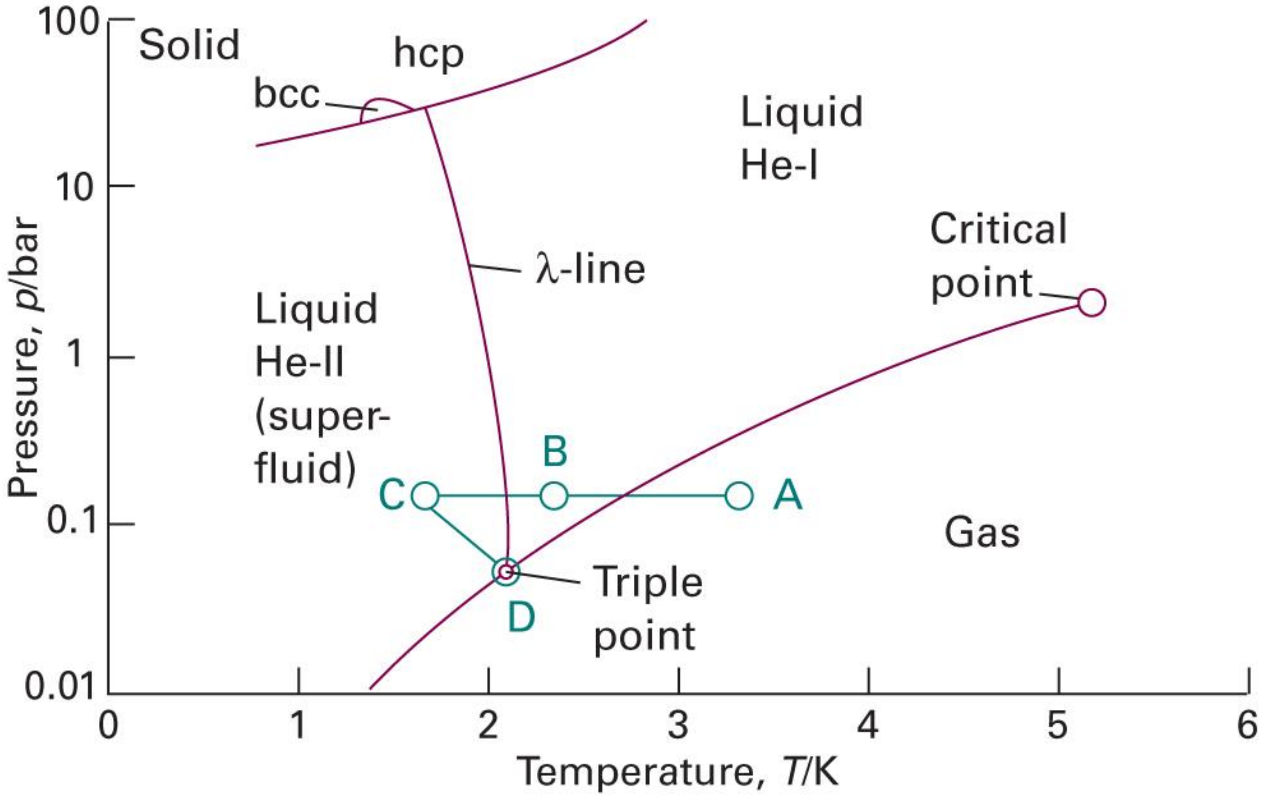






https://www.youtube.com/watch?v=UFGiBKYxHIw

Helium (^{4}He)



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figures 4A.11)

Phase Transition

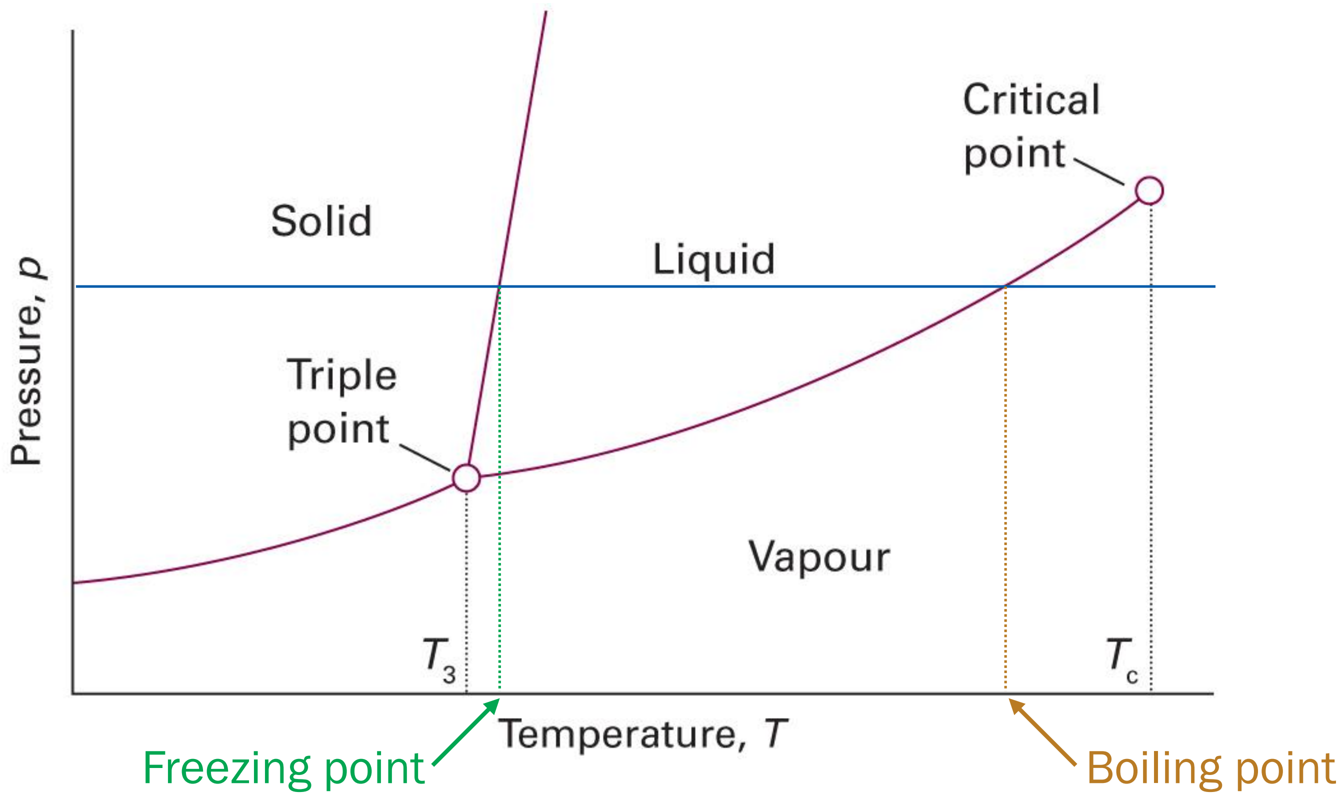
Spontaneous conversion

of one phase into another phase

- occurs at transition temperature T_{trs} (for given p)
- At T_{trs}, the two phases are at equilibrium

Topic 4A

Freezing and Boiling Points



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figures 4A.4)

Freezing and Boiling Points

- Normal boiling point T_b: b.p. at 1 atm
- Standard boiling point: b.p. at 1 bar
- e.g. water: T_b = 100 °C, standard b.p. = 99.6 °C

- Normal freezing point T_f: f.p. at 1 atm
- Standard freezing point: f.p. at 1 bar
- Usually the difference between the two is negligible

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Two Phases at Equilibrium

$$\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V$$

- Mechanistic equilibrium: $p_1 = p_2$
  - Conjugate extensive variable: $\mathrm{d}V$
- Thermal equilibrium: $T_1 = T_2$
  - Conjugate extensive variable: $\mathrm{d}S = \delta q / T$
- Diffusive equilibrium: ?
  - Conjugate extensive variable: $\mathrm{d}n$

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Chemical Potential

- Diffusive equilibrium: $\mu_1 = \mu_2$
  - Conjugate extensive variable: $\mathrm{d}n$

- (Modified) differentials of thermodynamic potentials

$$\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V + \mu\mathrm{d}n$$

$$\mathrm{d}H = T\mathrm{d}S + V\mathrm{d}p + \mu\mathrm{d}n$$

$$\mathrm{d}A = -S\mathrm{d}T - p\mathrm{d}V + \mu\mathrm{d}n$$

$$\mathrm{d}G = -S\mathrm{d}T + V\mathrm{d}p + \mu\mathrm{d}n$$

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3 Independent Variables

$$\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V + \mu\mathrm{d}n$$

This implies that $U = U(S, V, n)$, which leads to

$$\mathrm{d}U = \left(\frac{\partial U}{\partial S}\right)_{V,n} \mathrm{d}S + \left(\frac{\partial U}{\partial V}\right)_{S,n} \mathrm{d}V + \left(\frac{\partial U}{\partial n}\right)_{S,V} \mathrm{d}n$$

Comparison gives

$$\left(\frac{\partial U}{\partial S}\right)_{V,n} = T, \qquad \left(\frac{\partial U}{\partial V}\right)_{S,n} = -p, \qquad \left(\frac{\partial U}{\partial n}\right)_{S,V} = \mu$$

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Meaning of μ

$$\mathrm{d}G = -S\mathrm{d}T + V\mathrm{d}p + \mu\mathrm{d}n$$

This implies that $G = G(T, p, n)$.

G is extensive, and n is the only extensive variable. Thus,

$$G(T, p, n) = ng(T, p)$$

and

$$\left(\frac{\partial G}{\partial n}\right)_{T,p} = \left[\frac{\partial\{ng(T, p)\}}{\partial n}\right]_{T,p} = g(T, p)$$

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Meaning of μ

$$g(T,p) = \frac{G}{n} = \left(\frac{\partial G}{\partial n}\right)_{T,p}$$

However,

$$\mathrm{d}G = -S\mathrm{d}T + V\mathrm{d}p + \mu\mathrm{d}n$$

gives

$$\mu = \left(\frac{\partial G}{\partial n}\right)_{T,p} = g(T,p) = \frac{G}{n}$$

(molar Gibbs energy)

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Thermodynamic Potentials

$$\mu = G/n$$

can be written as

$$G = \mu n$$

From this equation, we can obtain

$$H = G + TS = \mu n + TS$$

$$A = G - pV = \mu n - pV$$

$$U = G + TS - pV = \mu n + TS - pV$$

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Gibbs-Duhem Equation

$$U = \mu n + TS - pV$$

Differentiating both sides,

$$\mathrm{d}U = \mu \mathrm{d}n + n\mathrm{d}\mu + T\mathrm{d}S + S\mathrm{d}T - p\mathrm{d}V - V\mathrm{d}p$$

but we know that

$$\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V + \mu \mathrm{d}n$$

Thus,

$S\mathrm{d}T - V\mathrm{d}p + n\mathrm{d}\mu = 0$ (Gibbs-Duhem equation)

$$\mathrm{d}\mu = -S_{\mathrm{m}}\mathrm{d}T + V_{\mathrm{m}}\mathrm{d}p$$

Two Systems in Contact

$$\mathrm{d}G = -S\mathrm{d}T + V\mathrm{d}p + \mu\mathrm{d}n$$

At constant T and p,

$$\mathrm{d}G = \mu\mathrm{d}n$$

For two systems in contact,

$$\mathrm{d}G_1 = \mu_1\mathrm{d}n_1$$

$$\mathrm{d}G_2 = \mu_2\mathrm{d}n_2$$

The total change in Gibbs energy is

$$\mathrm{d}G = \mathrm{d}G_1 + \mathrm{d}G_2 = \mu_1\mathrm{d}n_1 + \mu_2\mathrm{d}n_2$$

Diffusive Equilibrium

$$\mathrm{d}G = \mathrm{d}G_1 + \mathrm{d}G_2 = \mu_1 \mathrm{d}n_1 + \mu_2 \mathrm{d}n_2$$

If the two subsystems consist of the same type of particles and can exchange particles,

$$\mathrm{d}n_1 + \mathrm{d}n_2 = 0$$

Hence,

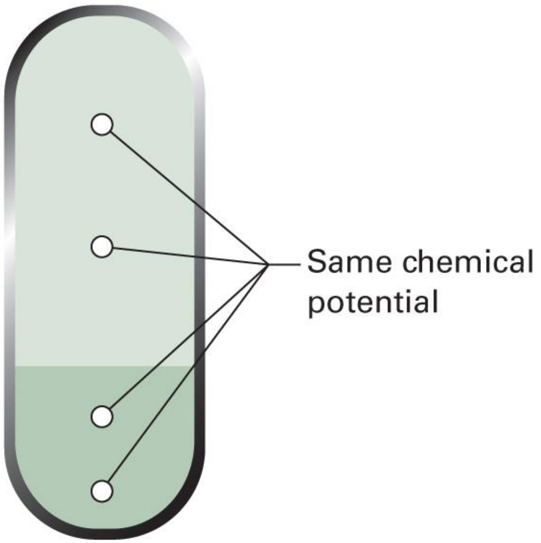
$$\mathrm{d}G = \mu_1 \mathrm{d}n_1 - \mu_2 \mathrm{d}n_1 = (\mu_1 - \mu_2)\mathrm{d}n_1$$

At equilibrium, $\mathrm{d}G = 0$, and

$$\mu_1 = \mu_2$$

Criteria for Phase Stability

At equilibrium, the chemical potential of a substance is the same in and throughout every phase in the system



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What If $\boldsymbol{\mu_1 \neq \mu_2}$?

$$\mathrm{d}G = \mu_1 \mathrm{d}n_1 - \mu_2 \mathrm{d}n_1 = (\mu_1 - \mu_2)\mathrm{d}n_1$$

- If $\mu_1 > \mu_2$, for $\mathrm{d}G < 0$ (spontaneous change),

$\mathrm{d}n_1 < 0$ and $\mathrm{d}n_2 = -\mathrm{d}n_1 > 0$,

which means that the particles move from system 1 to 2

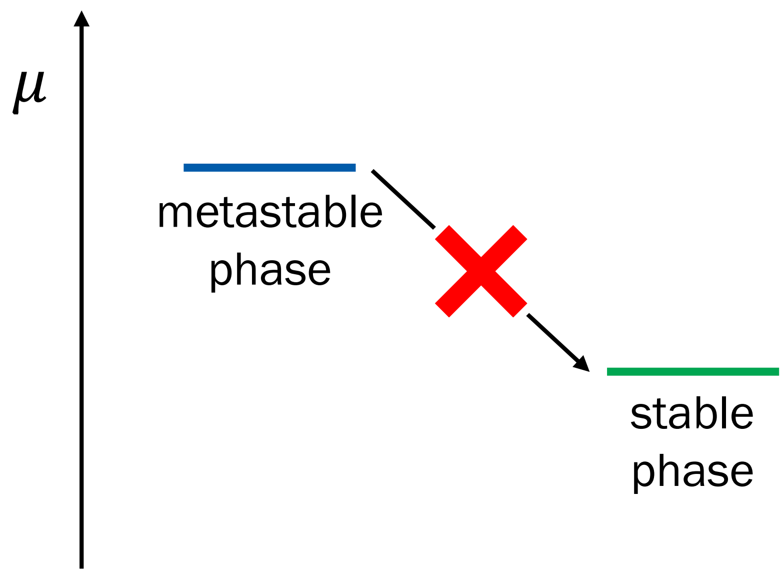
- If $\mu_1 < \mu_2$, for $\mathrm{d}G < 0$ (spontaneous change),

$\mathrm{d}n_1 > 0$ and $\mathrm{d}n_2 = -\mathrm{d}n_1 < 0$,

which means that the particles move from system 2 to 1

Metastable Phase

Thermodynamically unstable phases that persist because the transition is kinetically hindered



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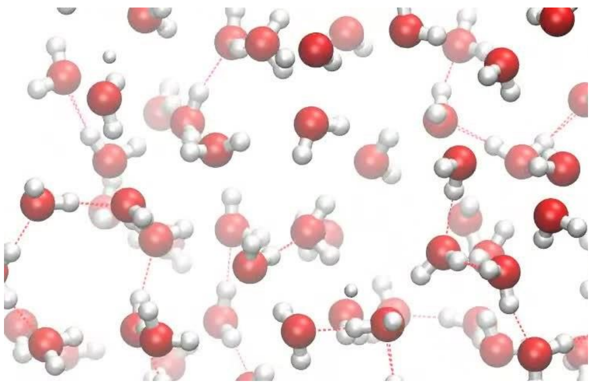
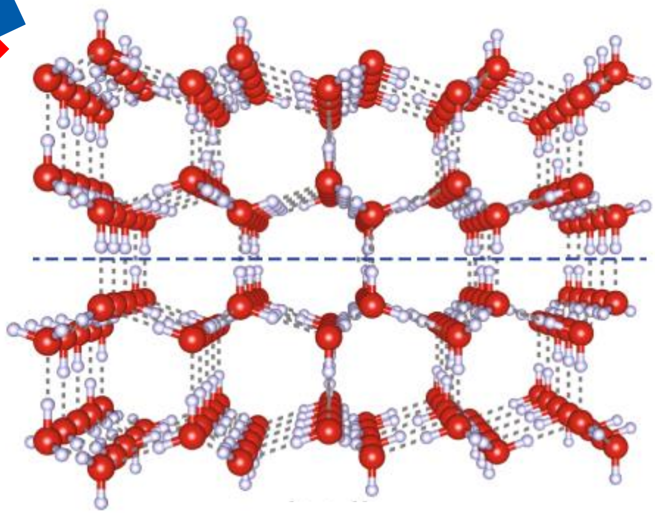
Supercooling



water (liquid)



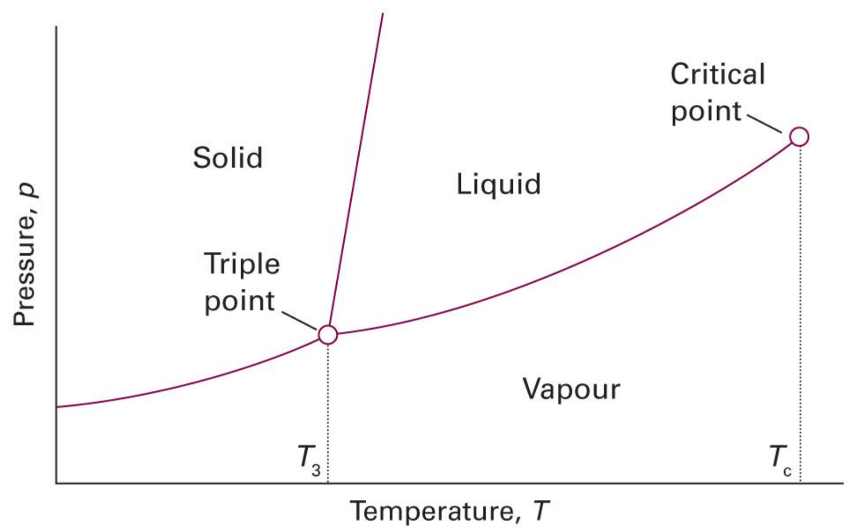
ice (solid)

https://www.youtube.com/watch?v=_9N-Y2CyYhM

(Image: https://commons.wikimedia.org/wiki/File:A_Molecular_Dynamics_Simulation_of_Liquid_Water_at_298_K.webm)

Phase Diagram

A diagram that shows the regions at which its various phases are thermodynamically stable



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figures 4A.4)

Phase Rule

"Can we have a quadruple point?"

- Phase rule: a general relation between
  - Number of degrees of freedom F
  - Number of components C
  - Number of phases at equilibrium P
- Use the criteria for phase stability (same μ)
  - Recall that $\mu = g(T, p)$

Single Component ($C = 1$)

- $P = 1$: both T and p can be independently varied

$$\therefore\ F = 2 \text{ (bivariant)}$$

- $P = 2$: $\mu_1(T,p) = \mu_2(T,p)$

$$\therefore\ F = 1 \text{ (univariant)}$$

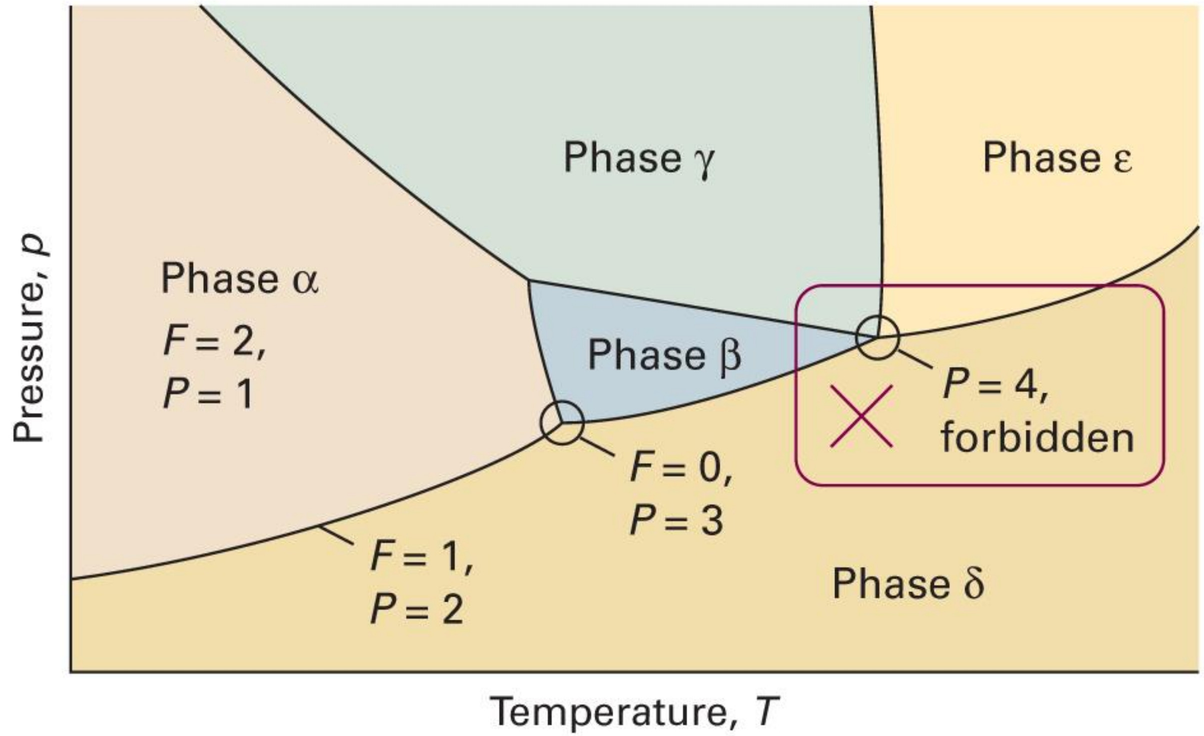
- $P = 3$: $\mu_1(T,p) = \mu_2(T,p)$ and $\mu_2(T,p) = \mu_3(T,p)$

$$\therefore\ F = 0 \text{ (invariant)}$$

- $P = 4$: two variables and three restrictions ⟶ impossible

In summary, $F = 3 - P$

Single Component ($C = 1$)



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figures 4A.7)

General Phase Rule

In general, the following rule holds:

$$F = C - P + 2$$

(Will be discussed in Chapter 7)

Let us focus on phase transitions

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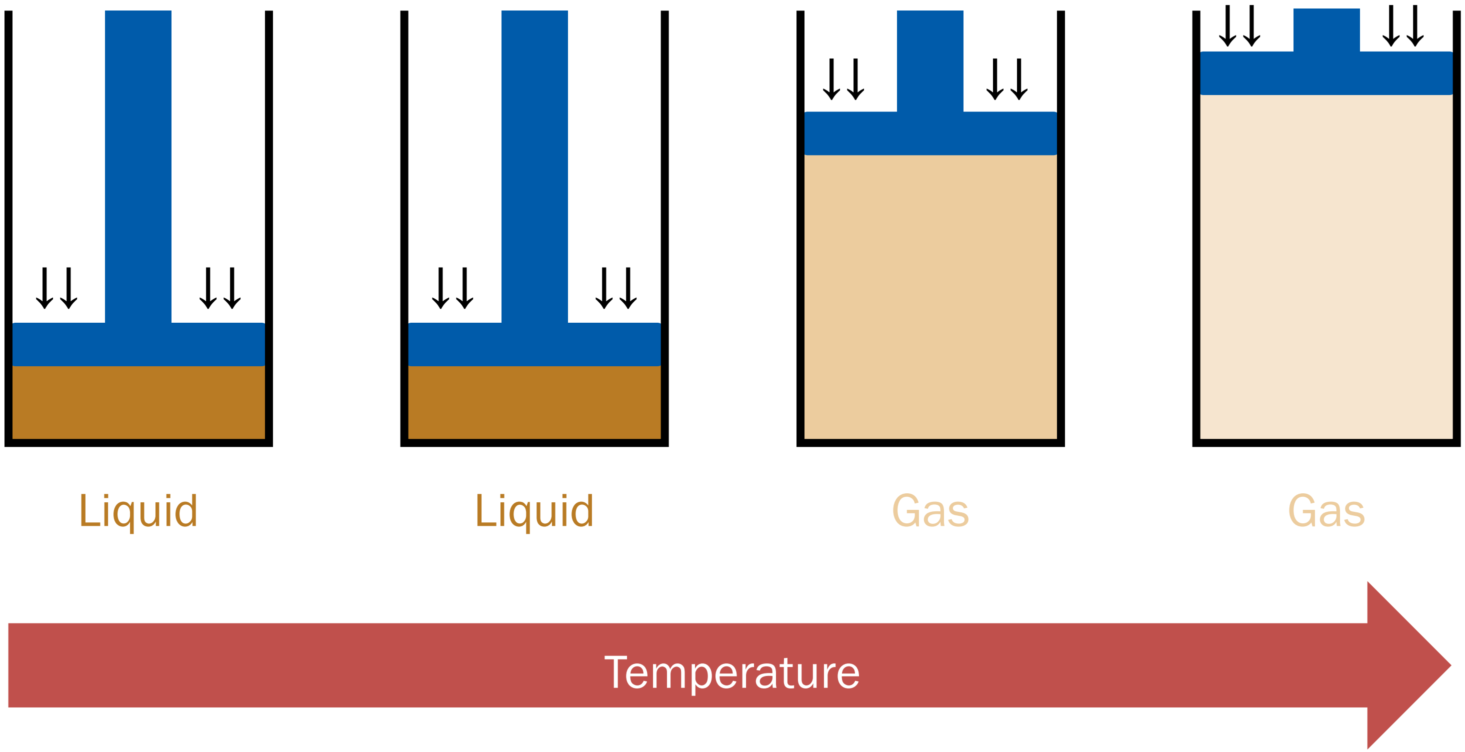
Liquid-Gas Transition 1

(Isobaric change)



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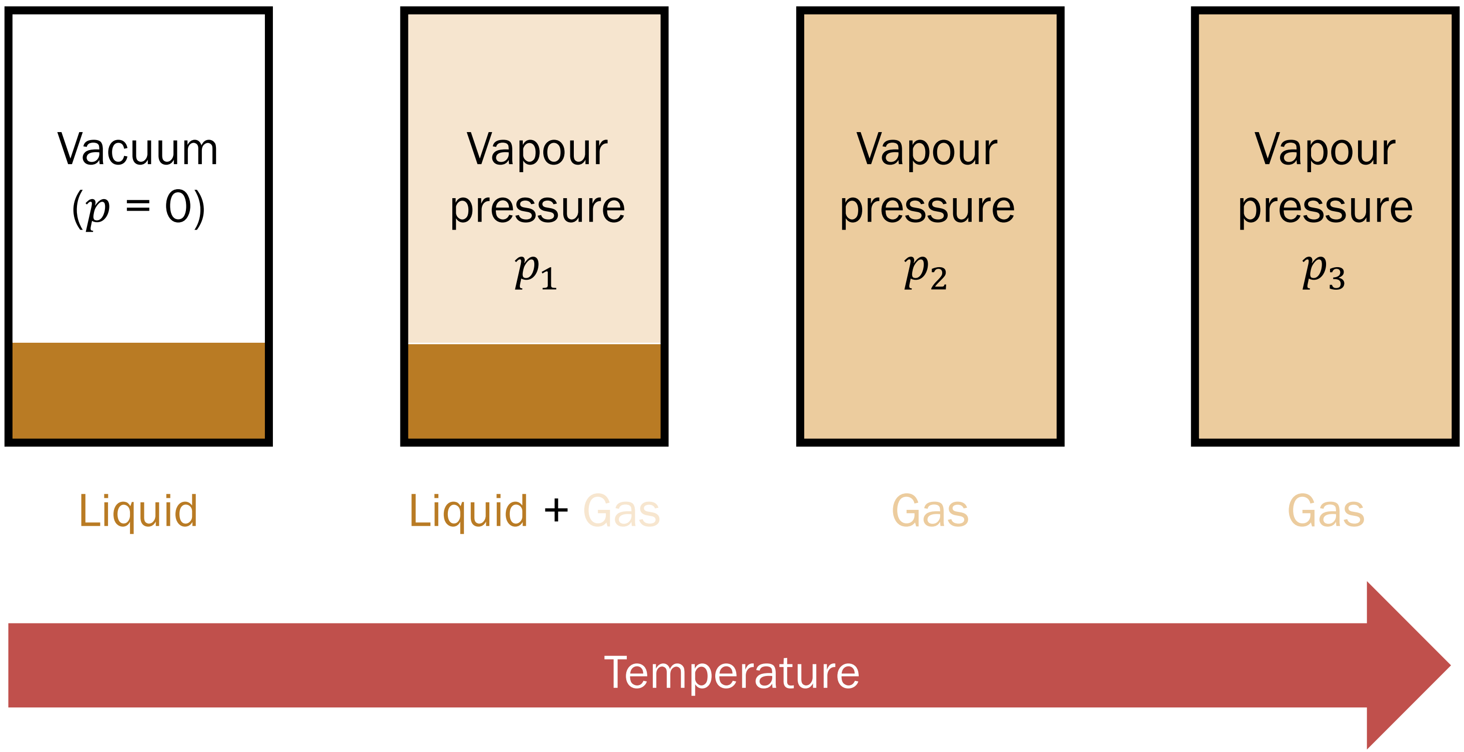
Liquid-Gas Transition 2

(Isochoric change)

$(p_1 < p_2 < p_3)$



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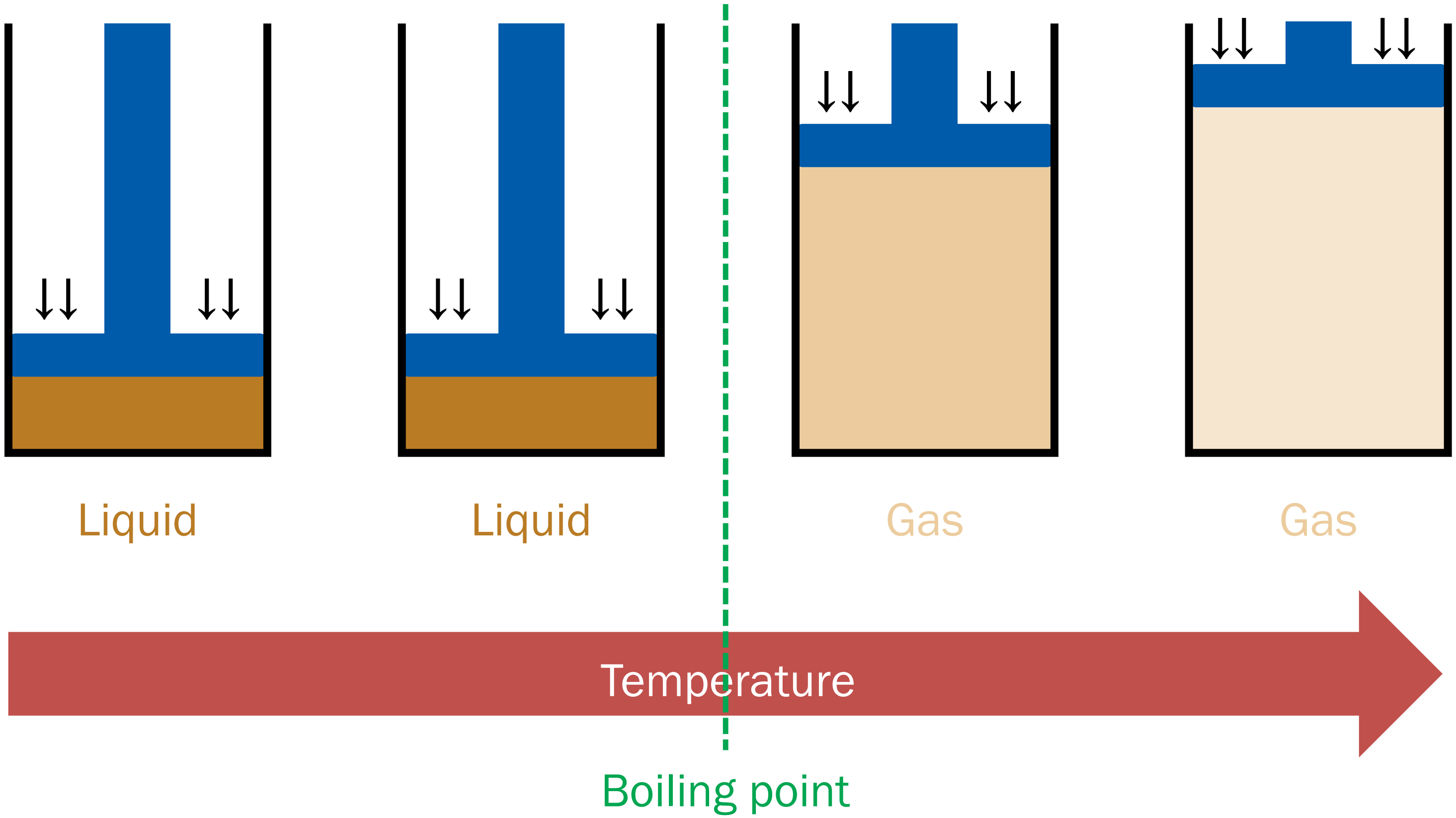
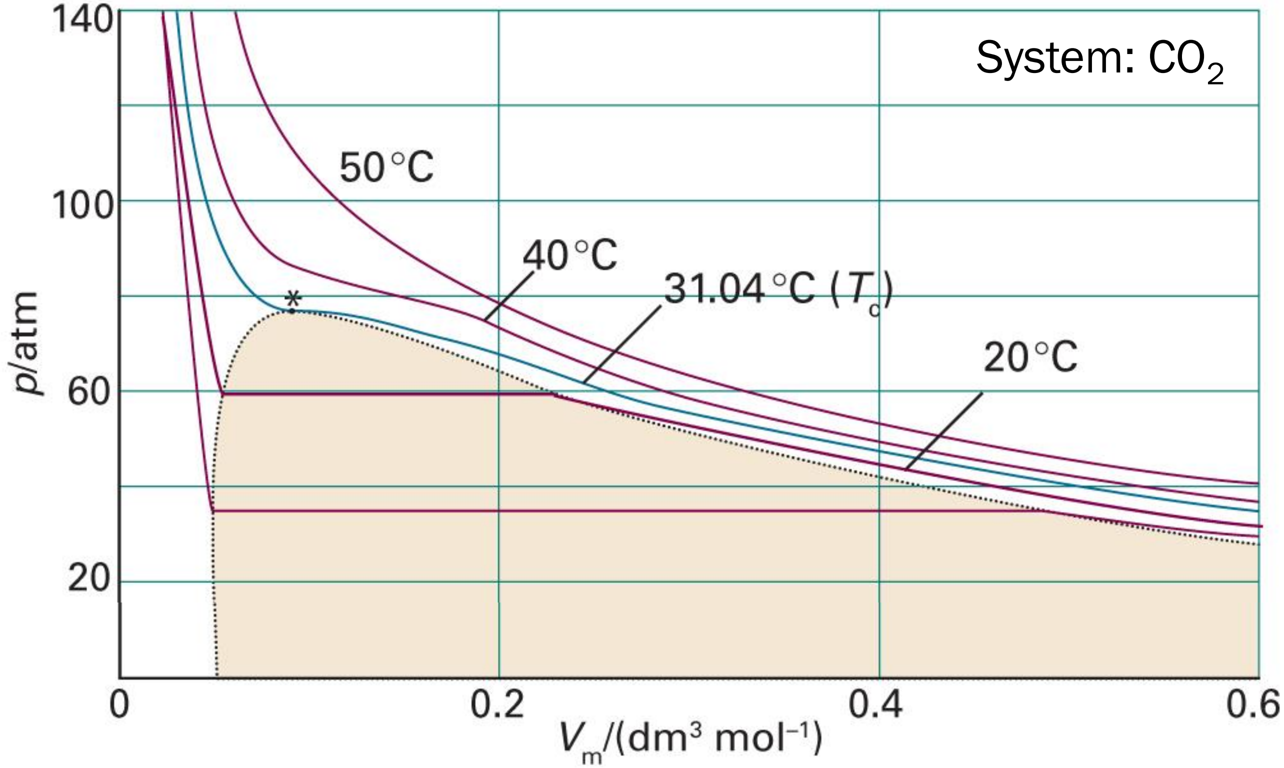
Vapor Pressure and Boiling

(Isobaric change)



Liquid-Gas Transition 3

(Isothermal change)



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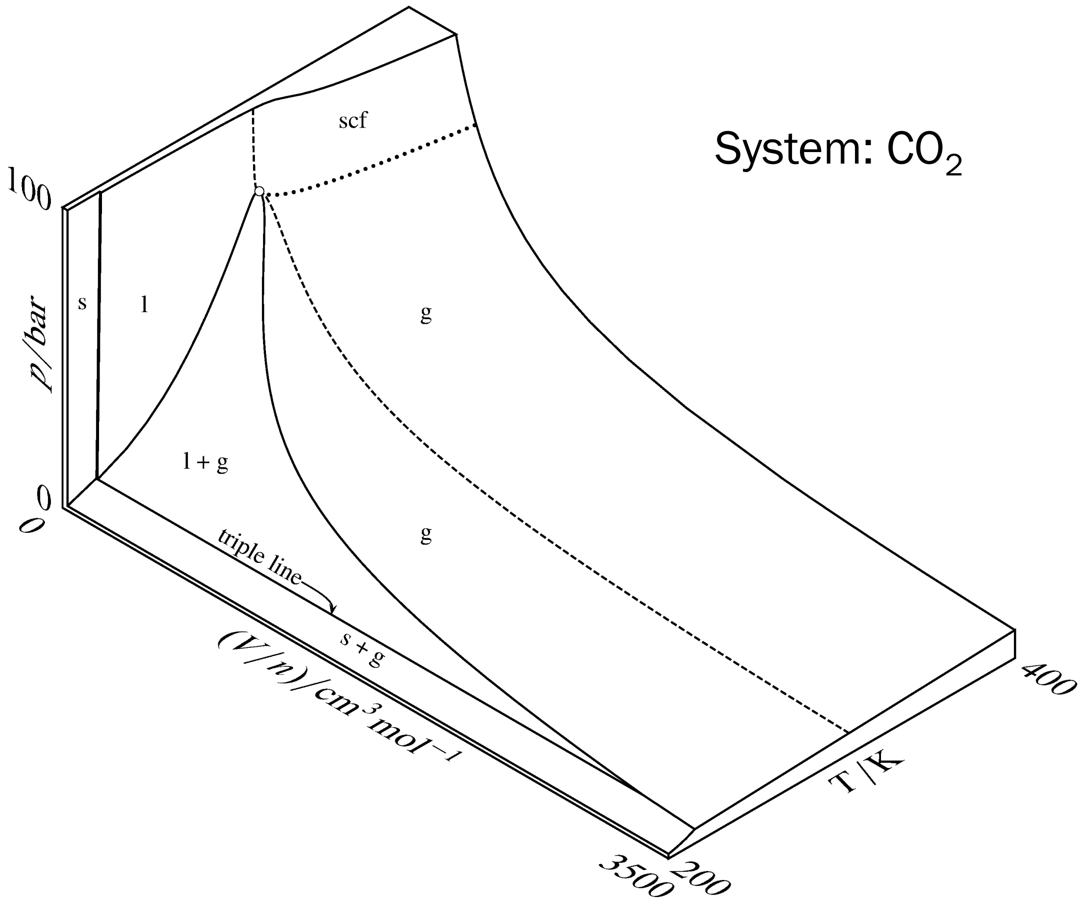
3-Dimensional Surface



System: CO_2

(Image: https://chem.libretexts.org/@go/page/20608)